

Finite Central Invisibility in Operational Reconstructions of Tensor Factorization and Horizon Entropy

June 12, 2026

Abstract

Operational reconstructions in algebraic quantum field theory and quantum information normally reconstruct the object that is supplied: a represented von Neumann algebra, a local net, a sector category, a channel tested by specified preparations and effects, or a Type II algebra with a trace normalization. This note proves a finite-central converse. On comparison classes closed under finite central direct sums, finite central refinements, and shared finite centers in proposed bipartitions, represented or retracted operational data determine full central support, central cardinality, tensor factorization, central channel dynamics, retained sector labels, and absolute finite-center entropy contributions exactly when the matching separating or calibrating hypothesis is supplied. If it is not supplied, explicit finite-central same-data pairs change the target invariant.

The result is a no-go theorem, not a new positive reconstruction of quantum gravity. It says that normal state/effect data factored through a quotient do not decide whether a nonzero finite central summand has been omitted; generated algebra correlations do not reconstruct an unquotiented spatial tensor product when multiplication has a kernel; and relative-entropy, recovery, modular-core, or crossed-product entropy data do not fix an absolute horizon entropy coefficient while a same-data finite-center or affine/trace residual has nonzero area density. Standard positive results—Tomita–Takesaki theory, Osterwalder–Schrader reconstruction, DHR/Doplicher–Roberts reconstruction, the split property, Fewster–Verch dynamical locality, process tomography, JLMS/OAQEC, and gravitational crossed-product entropy—are used as inputs and are left intact. A Lean 4 file formalizes the finite logical descent kernels; the analytic operator-algebraic reductions are the standard cited hypotheses.

Keywords. Algebraic quantum field theory; von Neumann algebras; finite centers; locally covariant quantum field theory; DHR sectors; split property; process tomography; modular crossed products; black-hole entropy.

MSC 2020. Primary 81T05, 46L10, 46L55; secondary 81P45, 83C57.

1 Question and Contribution

Many reconstruction arguments in quantum field theory and quantum information start from operational data: normal state values, effects, channels, relative entropies, modular flows, sector categories, or local probe readouts. The positive theorems then reconstruct the algebraic or geometric structure that was actually supplied. Tomita–Takesaki theory acts on a specified represented von Neumann algebra and weight. Osterwalder–Schrader reconstruction starts from reflection-positive Euclidean data after quotienting by the null space. DHR and Doplicher–Roberts reconstruction start from a specified sector category of a specified observable net. Locally covariant QFT and Fewster–Verch dynamical locality are formulated for a supplied functor and its relative Cauchy evolution. Process tomography is relative to specified preparations and effects; no-broadcasting constrains physical channels for noncommuting state families, not unsupplied commutative central coordinates. JLMS/OAQEC and crossed-product entropy use specified code algebras, represented algebras, traces, and normalizations [2, 3, 5, 6, 11, 12, 13, 14, 8, 9, 10, 19, 18, 21, 23, 27, 28, 29, 30, 31, 32, 33].

The question here is the converse finite-center question. Suppose the data have already passed through a represented quotient, a central-state retraction, a diagonal restriction, a generated algebra, a restricted input/effect family, or an entropy calibration that does not see all finite central directions. Can the data nevertheless reconstruct the full ambient central support, the number of finite central atoms, the unquotiented tensor product, or the absolute finite-center contribution to a horizon entropy coefficient?

The answer is no in the precise fiber sense below. A target invariant is reconstructible from a labelled data map exactly when it is constant on the data fibers. The theorem identifies the finite-central fibers for the standard operator-algebraic data maps named above. It also records the exact positive exits. Full support requires central support faithfulness. Central atoms and weights require central separation. Spatial tensor product coordinates require tested-kernel separation, and with full generated-algebra tomography this is injectivity of multiplication, supplied in AQFT by split or W^* -tensor-product independence. Finite central process dynamics require input spanning and output separation. Central sector labels require the supplied sector datum to retain those labels. Absolute entropy and horizon coefficients require trace, entropy, affine area, and geometric area-scale calibrations that kill same-data residuals.

Novelty claim. No individual ingredient in this note is advertised as new. Weak*-closed ideals of von Neumann algebras are central summands; normal representations have central support; normal functionals separate von Neumann algebras when enough of them are supplied; split inclusions give tensor-product independence; DHR/DR reconstruct from a supplied sector category; and Type II entropy constructions are relative to a specified algebra and trace. The claim is the joint finite-central descent criterion:

for the full-algebra targets listed below, represented, retracted, restricted, generated, or tested operational data determine the target if and only if the corresponding finite-central separator, tensor-kernel separator, sector-character datum, quotient/exclusion rule, or entropy/area calibration is supplied. If an existing paper already states this joint iff theorem, this note should be read as an application to operational spacetime and horizon-entropy reconstruction rather than as a new theorem.

2 Data Descent

Definition 1 (Reconstruction from labelled data). Let $\mathcal{X}$ be a comparison class, $D : \mathcal{X} \rightarrow \mathcal{D}$ a labelled data map, and $I : \mathcal{X} \rightarrow Y$ a target invariant. The target is reconstructible from D on $\mathcal{X}$ if there is a map $R : \mathcal{D} \rightarrow Y$ such that $I = R \circ D$.

Proposition 1 (Fiber criterion). *A target $I : \mathcal{X} \rightarrow Y$ is reconstructible from $D : \mathcal{X} \rightarrow \mathcal{D}$ if and only if I is constant on every fiber of D . Equivalently, no pair $X, Y \in \mathcal{X}$ has $D(X) = D(Y)$ but $I(X) \neq I(Y)$.*

Proof. If $I = R \circ D$, then $D(X) = D(Y)$ implies $I(X) = R(D(X)) = R(D(Y)) = I(Y)$. Conversely, if I is constant on data fibers, define $R(d) = I(X)$ for any X with $D(X) = d$. Fiber constancy makes this well-defined. $\square$

This elementary criterion is the only logical engine in the paper. The operator-algebraic work is to identify when ordinary operational data maps have finite central fibers on which a full-algebra target changes.

3 Main Finite-Central Descent Theorem

Theorem 1 (Finite-central operational descent). *Let $\mathcal{X}$ be a comparison class of von Neumann-algebraic local-net, locally covariant, code-algebra, or finite process models. Let $D : \mathcal{X} \rightarrow \mathcal{D}$ be a labelled operational data map built from normal state/effect values, channels tested by specified preparations and effects, represented modular or Osterwalder–Schrader data, DHR sector data, locally covariant relative-Cauchy-evolution or probe-measurement data, relative entropy or exact recovery data, crossed-product entropy data, or trace/area assignments.*

Assume that the finite central directions, when present, are read only through one of the following: a normal quotient, a represented GNS quotient, a central-state retraction, a diagonal subnet restriction, a generated algebra, a specified nonseparating preparation/effect family, a restricted sector functor, or a specified trace/area calibration. Assume also that the comparison class contains the corresponding finite central modifications unless an explicit data exit or exclusion exit is imposed.

For each full-algebra target in the left column, D reconstructs that target on the finite-central comparison class if and only if the matching exit in the right column is supplied.

<i>Target</i>	<i>Exact exit</i>
<i>Full central support of the supplied normal readouts</i>	<i>The readouts are faithful on the finite central positive cone.</i>
<i>Atoms and weights of a specified finite center</i>	<i>The restrictions of the supplied central readouts separate the center.</i>
<i>Variable finite cardinality, including the two-atom predicate</i>	<i>The data distinguish the corresponding cardinality fibers, or the cardinality is supplied as calibrated data.</i>
<i>Factor-valuedness of a retained local algebra</i>	<i>A factor, irreducibility, quotient, inadmissibility, or minimality hypothesis excludes the nontrivial finite center.</i>
<i>A proposed spatial tensor product for a bipartition</i>	<i>The tested pullbacks of multiplication have zero common kernel; with all normal generated-algebra correlations this is injectivity of multiplication, for example by split or W^*-tensor-product independence.</i>
<i>Finite central channel dynamics</i>	<i>Tested central preparations span the finite input algebra and tested central effects separate the output algebra.</i>
<i>Central-character labels in sector data</i>	<i>The supplied sector functor is faithful on the central-character fibers.</i>
<i>Absolute finite-center entropy, a horizon entropy-density coefficient, or the Newton coupling inferred from $S = A/(4G\hbar)$</i>	<i>The affine absolute calibration spans the claimed central state family, the trace calibration spans the central coordinates used by the entropy term, the geometric area scale is fixed, and every remaining same-data residual has zero limiting area density; or an explicit quotient/exclusion removes the residual.</i>

If the row exit is absent and the comparison class contains the corresponding finite central modification, then there are $X, Y \in \mathcal{X}$ with $D(X) = D(Y)$ and different values of the retained target. Hence no universal reconstruction rule from D can determine that target.

Proof. By Proposition 1, every row is a fiber computation.

For support, restrict the supplied normal positive functionals F to the finite center C . They rule out invisible central support exactly when every nonzero central projection $p \in C$ is detected by some $\varphi \in F$, i.e. $\varphi(p) > 0$. If such a projection is missed, the corner pMp can be retained or adjoined without changing any supplied readout. If no projection is missed, equality of readouts forbids a hidden positive central corner. This is precisely central faithfulness.

For atoms and weights of a specified finite center $C \simeq \ell_n^\infty$, the readouts induce a linear map

$$r_F : C \rightarrow \mathbb{C}^F, \quad r_F(c) = (\varphi(c))_{\varphi \in F}.$$

The central coordinates are reconstructed exactly when r_F is injective, or affinely injective on the normalized central state simplex. If r_F is not injective, two central observables or two central weights have identical labelled pairings. Across a comparison class in which n varies, the same fiber criterion says that finite cardinality, and in particular the predicate $n = 2$, is reconstructed only when the data distinguish the corresponding cardinality fibers.

Factor-valuedness is the assertion that the relevant center is trivial. A finite direct sum of factor fibers has a nontrivial finite center unless only one summand is present. Thus any data fiber that leaves finite cardinality or central separation undecided contains same-data candidates with different factor-valuedness. A factor, irreducibility, quotient, inadmissibility, or minimality hypothesis removes that pair from the comparison class.

For tensor factorization, let $\mathcal{A}, \mathcal{B} \subset \mathcal{N}(O)$ be commuting von Neumann algebras. Generated-algebra correlations evaluate an element $x \in \mathcal{A} \bar{\otimes} \mathcal{B}$ only through the normal multiplication representation

$$\mu : \mathcal{A} \bar{\otimes} \mathcal{B} \rightarrow \mathcal{A} \vee \mathcal{B}, \quad a \otimes b \mapsto ab.$$

With tested functionals $\Gamma \subset (\mathcal{A} \vee \mathcal{B})_*$, two tensor elements have the same generated-algebra data exactly when their difference lies in $\bigcap_{\gamma \in \Gamma} \ker(\gamma \circ \mu)$. The unquotiented spatial tensor coordinate therefore descends exactly when this common kernel is zero. If all normal functionals on $\mathcal{A} \vee \mathcal{B}$ are supplied, normal functionals separate points of the generated algebra, so the common kernel is $\ker \mu$. Split or W^* -tensor-product independence is the standard AQFT exit making this multiplication representation faithful in the cases considered here [15, 16, 17].

For finite central process data, the channel table consists of pairings $e(K\rho)$ between tested preparations and tested effects. The full finite central channel is determined exactly when tested preparations span the finite input vector space and tested effects separate the finite output vector space. If either condition fails, a finite classical channel can be perturbed in an untested direction without changing the displayed table.

For sector data, DHR/Doplicher–Roberts reconstruction applies to the sector category actually supplied [11, 12, 13, 14]. If finite central characters are forgotten by restriction to a diagonal subnet, those characters are not present in the supplied category. They are reconstructed exactly when the sector functor remains faithful on the central-character fibers.

For entropy and horizon coefficients, finite central residuals are explicit. Uniform q -fold multiplicity contributes $\log q$ for one unresolved finite center, or $m(\Sigma) \log q$ for $m(\Sigma)$ such centers on a cut Σ . Finite trace weights contribute logarithmic terms of the form $\sum_i p_i \log \lambda_i$. Relative-entropy and JLMS-type data determine absolute entropy or area potentials only modulo affine functions on the tested central state family. Therefore an absolute entropy term or density coefficient descends exactly when the supplied trace, state-family, affine, and geometric-area calibrations make every same-data residual vanish on the claimed family, or at least have zero limiting density after division

by area. If area units and $\hbar$ are fixed and $c > 0$, the map $c \mapsto (4\hbar c)^{-1}$ is injective, so the same criterion identifies the Newton coupling inferred from $S = A/(4G\hbar)$.

These row computations prove both directions. If the exit holds, the target is constant on finite-central fibers. If it fails and the comparison class contains the corresponding modification, one of the countermodels in the next section gives a same-data pair with different target value. $\square$

Closest prior results and novelty for Theorem 1. The support part is basic von Neumann algebra central-support theory [1, 2]. The tensor positive exit is the split/W*-independence literature [15, 16, 17]. The sector positive theorem is DHR/Doplicher–Roberts reconstruction from a supplied category [11, 12, 13, 14]. The locally covariant and measurement inputs are BFV and Fewster–Verch [8, 9, 10]. The entropy inputs are Araki/Petz/JLMS/OAQEC and gravitational crossed products [22, 23, 27, 28, 29, 30, 31, 32, 33]. The theorem’s claimed contribution is the simultaneous finite-central iff: these positive frameworks reconstruct full support, central cardinality, unquotiented tensor factors, retained sector labels, and absolute finite-center entropy contributions exactly when the corresponding finite-central exit is part of the supplied data or hypotheses.

4 Three Countermodels

4.1 Quotient-visible support and finite rank

Let $\mathcal{B}$ and $\mathcal{M}$ be nonzero von Neumann algebras. Put

$$\widehat{\mathcal{B}}_0 = \mathcal{B}, \quad \widehat{\mathcal{B}}_q = \mathcal{B} \oplus (\mathcal{M} \bar{\otimes} \ell^\infty(\{1, \dots, q\})), \quad q \geq 2,$$

and let $\pi_q : \widehat{\mathcal{B}}_q \rightarrow \mathcal{B}$ be the normal quotient onto the first summand, with $\pi_0 = \text{id}_{\mathcal{B}}$. Suppose every displayed datum is a labelled function of π_q , say

$$D_q = (\widetilde{D}_\lambda \circ \pi_q)_{\lambda \in \Lambda},$$

with fixed labels and codomains. Then the reported data are identical for $q = 0, 2, 3$. But for $q \geq 2$,

$$p_q = (0, 1_{\mathcal{M} \bar{\otimes} \ell_q^\infty})$$

is a nonzero central projection killed by every pulled-back readout, while no such complementary killed projection exists for $q = 0$. Thus quotient-visible data do not determine full ambient support, faithfulness on $\widehat{\mathcal{B}}_q$, or the interpretation of the killed kernel as absent, null, gauge, inadmissible, or physical. Comparing $q = 2$ and $q = 3$ also shows that the same data do not determine finite central cardinality or the two-atom predicate.

This example is not a claim that the extra summand is physically present. It is a no-go for deriving its absence from data that factor through π_q . A physical theory evades the witness by supplying representation faithfulness, full central tomography, a factor hypothesis, or an explicit quotient/gauge/null or inadmissibility principle.

4.2 Generated algebra versus spatial tensor product

Let $C = \ell^\infty(\{1, \dots, n\})$ with $n \geq 2$, and view C twice as commuting subalgebras of itself. The generated algebra is again C , while the candidate spatial tensor product is $C \bar{\otimes} C$. Multiplication acts by

$$\mu(e_i \otimes e_j) = \delta_{ij} e_i,$$

where e_i are the minimal central projections. Hence $e_i \otimes e_j \neq 0$ for $i \neq j$, but $\mu(e_i \otimes e_j) = 0$. Normal states on the generated algebra see only the diagonal image. Normal product states on $C \bar{\otimes} C$ separate the off-diagonal tensor atom.

Thus generated-algebra tomography is not the same as spatial tensor-product tomography. The exact repair is the usual one: injectivity of multiplication, for example from a split or W^* -tensor-product independence hypothesis.

4.3 Entropy density and Newton normalization

Let the displayed entropy data be insensitive to an unresolved finite center on a cut Σ , but let an entropy convention count $m(\Sigma)$ uniform q -point centers. The counted entropy changes by

$$\Delta S(\Sigma) = m(\Sigma) \log q.$$

If $A(\Sigma) \rightarrow \infty$ and

$$\frac{m(\Sigma)}{A(\Sigma)} \longrightarrow \beta,$$

then the entropy-density coefficient changes by $\beta \log q$. The same displayed represented, retracted, relative-entropy, modular, or crossed-product data can therefore correspond to two coefficients unless the finite-center counting convention is calibrated, quotiented, or shown to have zero density. After geometric area units and $\hbar$ are fixed, a positive coefficient c determines $G = (4\hbar c)^{-1}$. Thus a nonzero same-data residual in c is also a nonidentifiability of the Newton coupling inferred from the area law.

This is compatible with the positive Bekenstein–Hawking, JLMS/OAQEC, and crossed-product entropy results [24, 25, 27, 28, 29, 30, 31, 32, 33]. Those results supply or compute entropy for specified algebras and normalizations. The witness says only that a represented or retracted data map that does not include the finite central calibration cannot identify an absolute coefficient that counts it.

5 Tensor Factorization and Horizon Coefficient

Theorem 2 (Tensor factors and entropy coefficient from operational data). *Let $O \mapsto \mathcal{N}(O)$ be a Haag–Kastler net on a fixed spacetime, or the objectwise net of a locally covariant theory on a fixed globally hyperbolic spacetime. For each $i \in \mathcal{I}$, let $\mathcal{A}_i, \mathcal{B}_i \subset \mathcal{N}(O_i)$ be commuting von Neumann algebras and let*

$$\mu_i : \mathcal{A}_i \bar{\otimes} \mathcal{B}_i \rightarrow \mathcal{A}_i \vee \mathcal{B}_i, \quad a \otimes b \mapsto ab$$

be multiplication. Let $\Gamma_i \subset (\mathcal{A}_i \vee \mathcal{B}_i)_*$ be the supplied normal correlation functionals. Let E be a comparison class of entropy assignments on cuts Σ , with fixed geometric areas $A(\Sigma) > 0$, $A(\Sigma) \rightarrow \infty$, labelled entropy data $D_E : E \rightarrow \mathcal{D}_E$, and coefficient

$$c(X) = \lim_{\Sigma} \frac{S_X(\Sigma)}{A(\Sigma)}$$

when the limit exists.

For $x_i \in \mathcal{A}_i \bar{\otimes} \mathcal{B}_i$, define the labelled data

$$D((x_i), X) = ((\gamma(\mu_i(x_i)))_{i \in \mathcal{I}, \gamma \in \Gamma_i}, D_E(X)).$$

The target is the unquotiented tensor family $(x_i)_{i \in \mathcal{I}}$ together with $c(X)$. This target is reconstructible from D if and only if

$$\bigcap_{\gamma \in \Gamma_i} \ker(\gamma \circ \mu_i) = 0 \quad \text{for every } i \in \mathcal{I}$$

and

$$D_E(X) = D_E(Y) \implies \lim_{\Sigma} \frac{S_X(\Sigma) - S_Y(\Sigma)}{A(\Sigma)} = 0.$$

If $\Gamma_i = (\mathcal{A}_i \vee \mathcal{B}_i)_*$, the first condition is $\ker \mu_i = 0$. If area units and $\hbar$ are fixed and $c > 0$, the second condition is also the criterion for reconstructing the Newton coupling

$$G = \frac{1}{4\hbar c}.$$

Proof. Apply Proposition 1 coordinatewise. For fixed i , equality of generated-algebra correlation data for x_i and y_i means

$$\gamma(\mu_i(x_i - y_i)) = 0 \quad (\gamma \in \Gamma_i),$$

so $x_i - y_i$ lies in the common tested multiplication kernel. The unquotiented tensor coordinate is constant on the correlation fiber exactly when that kernel is zero. If all normal functionals on the generated algebra are supplied, normal functionals separate points, so the common kernel is $\ker \mu_i$.

The entropy coefficient is constant on D_E -fibers exactly when every same-data residual has zero limiting density, which is the second displayed condition. The map $c \mapsto (4\hbar c)^{-1}$ is injective for positive c , so the same criterion identifies the Newton coupling after the area scale and $\hbar$ are fixed. Failure of either condition gives a same-data pair by varying only the failed coordinate. $\square$

Closest prior results and novelty for Theorem 2. The tensor positive direction is the standard split/ W^* -independence exit [15, 16, 17]. The entropy positive inputs are Bekenstein–Hawking, FLM/JLMS/OAQEC, and Type II/crossed-product entropy for specified algebras and normalizations [24, 25, 26, 27, 28, 29, 30, 31, 32, 33]. The new packaging is the product fiber statement: generated-algebra operational correlations and entropy data identify the joint target only when the tensor-kernel condition and entropy-density calibration both hold. Either missing condition gives a same-data countermodel even if the other side is perfectly calibrated.

6 Relation to Positive Reconstruction Theorems

Positive input	What it supplies	Finite-central limitation isolated here
Tomita–Takesaki theory and crossed products	Modular flow and crossed product of a specified represented algebra and weight [2, 3, 4, 30].	They do not recover a finite central summand removed before representation; faithful full centers are carried along only when the full algebra is supplied.
DHR/Doplicher–Roberts reconstruction	Field algebra and compact gauge group from a supplied sector category [11, 12, 13, 14].	If diagonal restriction forgets central characters, those characters are not in the input category.
BFV/Fewster–Verch locally covariant QFT	Functorial dynamics, relative Cauchy evolution, dynamical locality, and local measurement schemes for a supplied functor and supplied probes [8, 9, 10].	Constant finite-center copies remain same-data after retraction unless the probes or object algebras separate the center.
Split inclusions	W^* -tensor-product independence for a supplied bipartition [15, 16, 17].	This is exactly the positive exit from the shared-center multiplication kernel; it is not implied by generated-algebra tomography alone.
Process tomography	Channel reconstruction relative to supplied preparations and effects [18, 19, 20].	Untested finite central input or output directions remain invisible.
No-broadcasting	Noncommuting mixed states cannot be broadcast by a universal physical channel [21].	The present obstruction is different: finite centers are commutative once supplied, but operational data that omit a central coordinate cannot infer which central coordinate or cardinality was present.

JLMS/OAQEC and crossed-product entropy	Entropy, recovery, or Type II algebra statements for specified algebras and normalizations [23, 27, 28, 29, 30, 31, 32, 33].	Absolute finite-center entropy terms and area coefficients require trace/area calibration or zero-density residuals.
--	--	--

The table is meant to prevent an inflated novelty claim. The note does not show that finite centers occur in nature and does not refute the positive theorems. It gives a test for proposed operational reconstructions: either the target factors through the supplied data, the hypotheses exclude the finite-central comparison pair, or the data contain the exact separator or calibration. If none of these is true, the three countermodels above supply a same-data pair with different target.

7 Assumption Ledger for Referees

This section is included to make the submission checkable without reading the longer manuscript. Each row separates a standard cited input from the new finite-central descent step and from the physical input that is deliberately not proved here.

Hypothesis used	Why it is standard	What this note does not prove
Normal quotients and represented support summands of von Neumann algebras.	Weak-* closed ideals are central summands, and normal representations have central support kernels [1, 2].	That a proposed physical theory admits no additional central summand, or that every quotient-zero direction is physically null.
Normal readouts restricted to a finite abelian center.	Finite-dimensional linear algebra: coordinate functionals separate ℓ_n^∞ , and positive functionals detect support exactly when every central atom is seen.	That a given operational principle supplies all such central readouts as physical effects.
Generated-algebra correlations for commuting subalgebras. Sector reconstruction from a supplied category.	Correlations on $\mathcal{A} \vee \mathcal{B}$ are functions of the multiplication image $\mu : \mathcal{A} \bar{\otimes} \mathcal{B} \rightarrow \mathcal{A} \vee \mathcal{B}$. DHR/Doplicher–Roberts reconstruction starts from the retained category of sectors [11, 12, 13, 14].	That μ is injective; this is the split/W*-independence exit when it holds. That a diagonal restriction has retained central-character labels it has already forgotten.

Locally covariant relative Cauchy evolution and local probes.	BFV and Fewster–Verch formulate these notions for a supplied functor and supplied probe couplings [8, 9, 10].	That every finite central direction has a source or induced probe effect separating it.
Process tomography.	Choi–Jamiolkowski tomography reconstructs a channel relative to an informationally complete input/effect domain [18, 19, 20].	That the admitted physical preparations and effects include the missing finite central directions.
Relative entropy, exact recovery, and crossed-product entropy.	These are positive results for specified algebras, channels, traces, and state families [22, 23, 27, 28, 29, 30, 31, 32, 33].	That an absolute finite-center entropy term or geometric area scale has been calibrated when the data include only relative or represented quantities.

The last column is not a defect in the cited theorems. It is the boundary of this note. A theory of quantum gravity may have a compelling physical reason for any of these exits: factor-valued local algebras, a gauge quotient, a superselection analysis retaining the relevant central characters, a split inclusion, or an absolute trace/area normalization. The point is that such a reason must appear as an assumption or theorem. It is not generated by data that were, by construction, constant on the finite-central comparison pair.

8 Novelty Boundary by Claim

Support and finite cardinality. The closest known ingredients are standard central support and quotient facts for von Neumann algebras. The new claim is not that central support exists, but that a quotient-visible operational package cannot reconstruct full ambient support, the kernel interpretation, or finite central rank unless the package contains a central separator or an exclusion rule. The $q = 0$ versus $q = 2$ comparison tests support totality; the $q = 2$ versus $q = 3$ comparison tests finite central cardinality and the two-atom predicate.

Tensor factorization. The closest positive result is the split property and W^* -tensor-product independence. The note’s contribution is the converse fiber formulation for operational data: generated-algebra correlations reconstruct only the multiplication image, so the unquotiented spatial tensor coordinate is identified exactly when the

tested multiplication kernel vanishes. This is a useful warning for operational reconstructions because full tomography of $\mathcal{A} \vee \mathcal{B}$ is still tomography of the quotient by $\ker \mu$, not of an arbitrary proposed tensor product.

Sectors and locally covariant data. DHR/DR and BFV/Fewster–Verch are positive frameworks for supplied data. The finite-center point is that restriction, retraction, or copied-center amplification can make distinct central-character or finite-center theories look identical to those supplied data. If a reconstruction theorem claims the full center anyway, it must include a sector-character separator, a probe effect/source separating the center, or an exclusion principle forbidding the comparison pair.

Broadcasting and copying. No-broadcasting theorems rule out a universal physical process broadcasting noncommuting mixed states. This note does not strengthen that theorem. Its finite centers are classical once included in the algebra; the obstruction is that represented, retracted, or restricted data may have omitted the central coordinate before any copying question is posed. The repair is therefore central separation or an exclusion principle, not a broadcasting channel.

Horizon entropy coefficient. Bekenstein–Hawking, FLM/JLMS, OAQEC, and gravitational crossed-product papers provide positive entropy formulas for specified geometric and algebraic normalizations. The finite-center descent statement is narrower: relative entropy, exact recovery, or a represented Type II algebra does not by itself identify an absolute finite-center counterterm if an affine, trace-weight, or counting residual has nonzero area density and is invisible to the data. Thus the exact one-quarter coefficient requires an identifiable calibration or a proof that every same-data residual has zero density.

Lean core. The Lean file is not a formal proof of AQFT. Its novelty status is modest but useful: it mechanically checks the finite logical fibers, quotient-factored no-go lemmas, diagonal tensor-kernel witness, and entropy-density descent criterion. This removes the risk that the finite part of the argument is a verbal sleight of hand. The infinite-dimensional reductions remain standard mathematical background, cited rather than formalized.

9 What Was Cut From the Long Manuscript

The longer manuscript contains many variants: modular-core refinements, OS specializations, process-tomography examples, local-probe protocols, sector-character wrappers, affine/JLMS calibration variants, and several Lean wrappers. They are omitted here unless they are needed for one of the two main statements or for the three countermodels. This is intentional. A referee for the short note should have to check only the following chain:

1. the fiber criterion in Proposition 1;
2. the standard operator-algebra facts cited in the assumption ledger;
3. the row-by-row finite-center calculation in Theorem 1;
4. the three same-data witnesses;
5. the coordinatewise tensor/entropy-density criterion in Theorem 2;
6. the Lean file as a finite logical sanity check, not as evidence for any analytic AQFT premise.

Everything else is deferred to the longer manuscript or to future work. In particular, this note does not attempt to prove a microscopic origin of a finite center, does not prove that physical local algebras fail to be factors, does not derive the Bekenstein–Hawking coefficient, and does not classify all locally covariant theories. It gives a finite-central obstruction that a positive reconstruction theorem must evade explicitly.

10 Lean Core

The accompanying Lean 4 file

`formalization/FiniteCenterSupport.lean`

does not formalize AQFT, Tomita–Takesaki theory, DHR sectors, or crossed products. It formalizes the finite logical kernels that remain after the standard analytic hypotheses reduce the question to finite central fibers. The most relevant named statements are:

`quotientFactoredInvariantNoGoCore,`
`quotientFactoredPropertyNoGoCore,`
`quotientVisibleUniversalNoGoCore,`
`finiteCenterOperationalCompletenessCore,`
`finiteCenterLocalNetClassificationCore,`
`localNetFamilyEntropyDescentCore,`
`refereeTensorEntropyCriterionCore,`
`universalFiniteCentralWitnessesCore.`

These lemmas mechanically check the fiber criterion, quotient-factored nonreconstruction, finite support coverage, finite coordinate separation, diagonal multiplication kernels, finite process-table completeness, and entropy-density descent. The file therefore verifies the finite part of the paper’s logic while leaving the infinite-dimensional operator-algebraic inputs to the cited literature.

11 Conclusion

The finite-central obstruction is small but sharp. Operational data that have already passed through a quotient, representation, retraction, generated algebra, restricted test family, or incomplete calibration cannot reconstruct a full-algebra target that changes inside the corresponding data fiber. For emergent-spacetime reconstructions, this means that tensor factorization and the exact horizon entropy coefficient require identifiable extra hypotheses: split/tensor independence for the former, and trace/area/entropy calibration with zero same-data density residual for the latter. Those hypotheses may be physically correct in a given theory. The theorem says only that they must be present and checkable; they cannot be replaced by operational data that have already forgotten the finite center.

References

- [1] S. Sakai, *C*-Algebras and W*-Algebras*, Springer, 1971.
- [2] M. Takesaki, *Theory of Operator Algebras I*, Springer, 1979.
- [3] M. Takesaki, *Theory of Operator Algebras II*, Springer, 2003.
- [4] M. Takesaki, “Duality for crossed products and the structure of von Neumann algebras of type III,” *Acta Math.* **131** (1973) 249–310, doi:10.1007/BF02392041.
- [5] K. Osterwalder and R. Schrader, “Axioms for Euclidean Green’s functions,” *Commun. Math. Phys.* **31** (1973) 83–112, doi:10.1007/BF01645738.
- [6] K. Osterwalder and R. Schrader, “Axioms for Euclidean Green’s functions II,” *Commun. Math. Phys.* **42** (1975) 281–305, doi:10.1007/BF01608978.
- [7] R. Haag and D. Kastler, “An algebraic approach to quantum field theory,” *J. Math. Phys.* **5** (1964) 848–861, doi:10.1063/1.1704187.
- [8] R. Brunetti, K. Fredenhagen and R. Verch, “The generally covariant locality principle – A new paradigm for local quantum physics,” *Commun. Math. Phys.* **237** (2003) 31–68, arXiv:math-ph/0112041, doi:10.1007/s00220-003-0815-7.
- [9] C. J. Fewster and R. Verch, “Dynamical locality and covariance: What makes a physical theory the same in all spacetimes?” *Ann. Henri Poincaré* **13** (2012) 1613–1674, arXiv:1106.4785, doi:10.1007/s00023-012-0165-0.
- [10] C. J. Fewster and R. Verch, “Quantum fields and local measurements,” *Commun. Math. Phys.* **378** (2020) 851–889, arXiv:1810.06512, doi:10.1007/s00220-020-03800-6.

- [11] S. Doplicher, R. Haag and J. E. Roberts, “Fields, observables and gauge transformations I,” *Commun. Math. Phys.* **13** (1969) 1–23, doi:10.1007/BF01645267.
- [12] S. Doplicher, R. Haag and J. E. Roberts, “Local observables and particle statistics I,” *Commun. Math. Phys.* **23** (1971) 199–230, doi:10.1007/BF01877742.
- [13] S. Doplicher and J. E. Roberts, “A new duality theory for compact groups,” *Invent. Math.* **98** (1989) 157–218, doi:10.1007/BF01388849.
- [14] S. Doplicher and J. E. Roberts, “Why there is a field algebra with a compact gauge group describing the superselection structure in particle physics,” *Commun. Math. Phys.* **131** (1990) 51–107, doi:10.1007/BF02097680.
- [15] S. Doplicher and R. Longo, “Standard and split inclusions of von Neumann algebras,” *Invent. Math.* **75** (1984) 493–536, doi:10.1007/BF01388641.
- [16] D. Buchholz and E. H. Wichmann, “Causal independence and the energy-level density of states in local quantum field theory,” *Commun. Math. Phys.* **106** (1986) 321–344, doi:10.1007/BF01454978.
- [17] R. F. Werner, “Local preparability of states and the split property in quantum field theory,” *Lett. Math. Phys.* **13** (1987) 325–329, doi:10.1007/BF00401161.
- [18] A. Jamiołkowski, “Linear transformations which preserve trace and positive semidefiniteness of operators,” *Rep. Math. Phys.* **3** (1972) 275–278, doi:10.1016/0034-4877(72)90011-0.
- [19] M.-D. Choi, “Completely positive linear maps on complex matrices,” *Linear Algebra Appl.* **10** (1975) 285–290, doi:10.1016/0024-3795(75)90075-0.
- [20] I. L. Chuang and M. A. Nielsen, “Prescription for experimental determination of the dynamics of a quantum black box,” *J. Mod. Opt.* **44** (1997) 2455–2467, arXiv:quant-ph/9610001, doi:10.1080/09500349708231894.
- [21] H. Barnum, C. M. Caves, C. A. Fuchs, R. Jozsa and B. Schumacher, “Noncommuting mixed states cannot be broadcast,” *Phys. Rev. Lett.* **76** (1996) 2818–2821, arXiv:quant-ph/9511010, doi:10.1103/PhysRevLett.76.2818.
- [22] H. Araki, “Relative entropy of states of von Neumann algebras,” *Publ. Res. Inst. Math. Sci.* **11** (1976) 809–833, doi:10.2977/prims/1195191148.
- [23] D. Petz, “Sufficient subalgebras and the relative entropy of states of a von Neumann algebra,” *Commun. Math. Phys.* **105** (1986) 123–131, doi:10.1007/BF01212345.
- [24] J. D. Bekenstein, “Black holes and entropy,” *Phys. Rev. D* **7** (1973) 2333–2346, doi:10.1103/PhysRevD.7.2333.

- [25] S. W. Hawking, “Particle creation by black holes,” *Commun. Math. Phys.* **43** (1975) 199–220, doi:10.1007/BF02345020.
- [26] T. Faulkner, A. Lewkowycz and J. Maldacena, “Quantum corrections to holographic entanglement entropy,” *JHEP* **11** (2013) 074, arXiv:1307.2892, doi:10.1007/JHEP11(2013)074.
- [27] D. L. Jafferis, A. Lewkowycz, J. Maldacena and S. J. Suh, “Relative entropy equals bulk relative entropy,” *JHEP* **06** (2016) 004, arXiv:1512.06431, doi:10.1007/JHEP06(2016)004.
- [28] D. Harlow, “The Ryu–Takayanagi formula from quantum error correction,” *Commun. Math. Phys.* **354** (2017) 865–912, arXiv:1607.03901, doi:10.1007/s00220-017-2904-z.
- [29] X. Dong, D. Harlow and A. C. Wall, “Reconstruction of bulk operators within the entanglement wedge in gauge-gravity duality,” *Phys. Rev. Lett.* **117** (2016) 021601, arXiv:1601.05416, doi:10.1103/PhysRevLett.117.021601.
- [30] E. Witten, “Gravity and the crossed product,” *JHEP* **10** (2022) 008, arXiv:2112.12828, doi:10.1007/JHEP10(2022)008.
- [31] R. Longo and E. Witten, “A note on continuous entropy,” arXiv:2202.03357.
- [32] V. Chandrasekaran, R. Longo, G. Penington and E. Witten, “An algebra of observables for de Sitter space,” *JHEP* **02** (2023) 082, arXiv:2206.10780, doi:10.1007/JHEP02(2023)082.
- [33] V. Chandrasekaran, G. Penington and E. Witten, “Large N algebras and generalized entropy,” *JHEP* **04** (2023) 009, arXiv:2209.10454, doi:10.1007/JHEP04(2023)009.